

TimesFM (Foundation Model) Testing

Notebooks 5, 7 and 8 **train** models on this series. This notebook tests **TimesFM**, a pretrained time-series *foundation model* from Google Research: a decoder-only transformer trained on a very large corpus of real and synthetic series, which forecasts a new series **zero-shot**, without any training on it. The model only reads a *context* (the recent history) and continues it.

With no weights to fit, what can still be chosen is **what the model reads**:

Choice	Options
Context window ("training" cut)	all history, the last 14 / 12 / 10 / 8 / 6 / 4 years, post-COVID (as in notebooks 5 and 7)
COVID rows	keep , remove 2020–2022 (as in notebook 7), or mask the pandemic months as missing
Target processing	none , log , seasonal_log_diff
Auxiliary features (covariates)	none, calendar, COVID flags, calendar + COVID flags
Symmetric averaging	off / on (an inference option of the model)

TimesFM is used on the **aggregated series only** (the total monthly arrivals of notebooks 5 and 7), not as a global model on the panel of notebook 8.

The protocol is the one of notebooks 5, 7 and 8, so the results are directly comparable: the same four validation origins, the **mean seasonal MASE** over them as the selection metric, the same baselines, MLflow logging (`eval_split = validation`) and p95 heatmaps. The test set is never read here. TimesFM also returns **quantiles**, so the 80% forecast interval is evaluated too.

Setup

```
In [1]: import itertools
import logging
import os
import sys
import time
import warnings
from concurrent.futures import ThreadPoolExecutor
from typing import NamedTuple

import matplotlib.pyplot as plt
import mlflow
import numpy as np
```

```

import pandas as pd
import seaborn as sns
from dotenv import load_dotenv
from IPython.display import display
from sklearn.metrics import r2_score
from timesfm3 import TimesFM3Forecaster
from tqdm import tqdm

load_dotenv()

ROOT_PATH = os.path.dirname(os.getcwd())
DATA_FOLDER_PATH = os.path.join(ROOT_PATH, 'data', 'processed')

COVID_START = pd.Timestamp('2020-04-01') # first pandemic month: April 2020
COVID_END = pd.Timestamp('2021-12-31')
POST_COVID_START = pd.Timestamp('2022-01-01')
COVID_RECOVERY_END = pd.Timestamp('2022-06-30')

TARGET = 'arrival_count'
FREQ = 'ME'
SEASONAL_PERIOD = 12

sns.set_palette('colorblind')
pd.set_option('display.max_columns', 40)
pd.set_option('display.width', 200)
logging.getLogger('mlflow').setLevel(logging.ERROR)
warnings.filterwarnings('ignore', category=FutureWarning)
mlflow.set_tracking_uri(os.getenv('MLFLOW_TRACKING_URI'))
EXPERIMENT_NAME = 'Brazil Tourism Forecasting TimesFM'
experiment = mlflow.set_experiment(EXPERIMENT_NAME)

```

The Model and What It Supports

Das, A., Kong, W., Sen, R. & Zhou, Y. *A decoder-only foundation model for time-series forecasting*. ICML 2024 — [arXiv:2310.10688](https://arxiv.org/abs/2310.10688); code and weights: [google-research/timesfm](https://github.com/google-research/timesfm) (code Apache-2.0).

The notebook uses **TimesFM 3.0** (`google/timesfm-3.0-pytorch` , package `timesfm[torch]` , module `timesfm3`), the latest release (August 2026). Its pretrained weights are distributed under the `timesfm-non-commercial-license-v1.0` : research and non-commercial use only, which covers this thesis. TimesFM 2.5 (Apache-2.0 weights) would be the choice for a commercial deployment.

What the model supports, and how each capability is used here:

Capability (<code>TimesFM3Forecaster.predict</code>)	Used here
Univariate forecasting from a context of up to 15,360 points	✓ the total series; the context length is the "training cut"

Capability (<code>TimesFM3Forecaster.predict</code>)	Used here
Past-and-future covariates (<code>past_future_covariates</code>): numeric series known over the context and the horizon	✓ calendar terms and COVID flags, known in advance for any month
Past-only covariates (<code>past_only_covariates</code>): numeric series known only over the context	X they would be other observed series (e.g. the components of notebook 8); the aggregated-only scope leaves them out
Multivariate targets (up to 32 jointly forecast series)	X same reason (notebook 8's panel)
Missing values (<code>NaN</code>) in the context, linearly interpolated inside the model	✓ the <code>mask</code> COVID option
Quantiles 0.1 ... 0.9 (<code>return_quantiles=True</code>); the point forecast is the median	✓ the 80% interval (0.1–0.9) is scored
<code>use_symmetric_averaging</code> : also forecasts the sign-flipped series and averages the two	✓ grid option
<code>make_positive</code> : clamps the forecast at zero	✓ always on for arrivals on the original scale
<code>use_znorm</code> : z-normalizes each series before the model	X the model already normalizes its input internally; on this series it gave identical forecasts

Covariates are numeric only; the categorical month is passed as its sine and cosine.

The model runs on the CPU in a fraction of a second per forecast, so the whole grid runs sequentially in one process (no parallel workers, which would each load their own copy of the model).

```
In [2]: MODEL_ID = 'google/timesfm-3.0-pytorch'
        HORIZON = 12

        model = TimesFM3Forecaster.from_pretrained(MODEL_ID)
        QUANTILES = list(model.config.quantiles)
        Q_LOW, Q_HIGH = QUANTILES.index(0.1), QUANTILES.index(0.9)
        print(
            f'{MODEL_ID} on {model.device} | max context {model.global_context:,} | '
            f'quantiles {QUANTILES}'
        )
```

Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster downloads.

google/timesfm-3.0-pytorch on cpu | max context 15,360 | quantiles [0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9]

```
In [3]: # Book helper credited to: Joseph, M. & Tackes, J. "Modern Time Series Forecasting
        # Python", 2nd ed. (Packt, 2024) – MIT License
        # github.com/PacktPublishing/Modern-Time-Series-Forecasting-with-Python-2E
        MTSF_REPO_PATH = os.getenv(
            'MTSF_REPO_PATH',
```

```

    os.path.join(
        os.path.dirname(ROOT_PATH), 'Modern-Time-Series-Forecasting-with-Python-2E'
    ),
)
if MTSF_REPO_PATH not in sys.path:
    sys.path.insert(0, MTSF_REPO_PATH)

from src.forecasting.ml_forecasting import calculate_metrics # noqa: E402

```

1. Data and Validation Origins

Monthly totals from `train.parquet` and `val.parquet` (the test set is never read), and the same validation origins as notebooks 5, 7 and 8: the end of training (its window is `val.parquet`) and one year earlier, plus the two most recent yearly origins whose 12-month window ends before the pandemic. They are counted back from the end of training, so they roll forward with the data. At each origin the model only reads the history up to that origin.

```

In [4]: def load_monthly(file_name):
        raw = pd.read_parquet(os.path.join(DATA_FOLDER_PATH, file_name))
        return raw.set_index('date')[TARGET].resample(FREQ).sum()

train_series = load_monthly('train.parquet')
val_series = load_monthly('val.parquet')
TRAIN_END = train_series.index[-1]
assert val_series.index[0] == TRAIN_END + pd.offsets.MonthEnd(1)
assert len(val_series) == HORIZON
# train + validation: every origin is cut from this history (never the test set)
full_series = pd.concat([train_series, val_series])

N_PRE_COVID_ORIGINS = 2
N_POST_COVID_ORIGINS = 2

def validation_window(origin):
    """First and last month forecast from `origin` (the last month the model sees).
    return origin + pd.offsets.MonthEnd(1), origin + pd.offsets.MonthEnd(HORIZON)

candidates = [TRAIN_END] + [
    TRAIN_END - pd.offsets.MonthEnd(12 * k) for k in range(1, len(train_series) //
]
post_covid_origins = [o for o in candidates if validation_window(o)[0] > COVID_END]
                    :N_POST_COVID_ORIGINS
]
pre_covid_origins = [o for o in candidates if validation_window(o)[1] < COVID_START]
                    :N_PRE_COVID_ORIGINS
]
ORIGINS = sorted(pre_covid_origins + post_covid_origins)
LATEST_ORIGIN = TRAIN_END # its validation window is val.parquet
assert LATEST_ORIGIN in ORIGINS

```

```

def origin_label(origin):
    return f'{origin:%Y-%m}'

def val_actual_at(origin):
    start, end = validation_window(origin)
    return full_series.loc[start:end]

pd.DataFrame([
    {
        'origin': origin_label(origin),
        'phase': 'pre-COVID' if origin < COVID_START else 'post-COVID',
        'validation window': f'{start:%b %Y} → {end:%b %Y}',
        'history months': len(full_series.loc[:origin]),
    }
    for origin in ORIGINS
    for start, end in [validation_window(origin)]
])

```

Out[4]:

	origin	phase	validation window	history months
0	2017-08	pre-COVID	Sep 2017 → Aug 2018	344
1	2018-08	pre-COVID	Sep 2018 → Aug 2019	356
2	2023-08	post-COVID	Sep 2023 → Aug 2024	416
3	2024-08	post-COVID	Sep 2024 → Aug 2025	428

2. Building the Context

A zero-shot model is steered only through its input. Each configuration turns the history of one origin into a context in four steps:

1. **Context window** (the "training" cut of the other notebooks): the last N years before the origin, all history, or January 2022 onwards (`post_covid`).
2. **COVID rows.**
 - `keep` : the series as it is.
 - `remove` : the years 2020–2022 are cut out and December 2019 is followed by January 2023 (three full years, so the months stay aligned; notebook 7).
 - `mask` : the pandemic and rebound months (April 2020 – December 2022) stay in place as **missing values**; TimesFM interpolates them linearly. The calendar positions are kept and the model sees no collapse. The mask is applied **after** the target processing, so the seasonal difference of 2023 is still computed from real 2022 values.

Before the pandemic there is nothing to remove or mask, so at the pre-COVID origins both are the same as `keep`. `post_covid` only runs with `keep`.

3. Target processing, fitted on the context:

| Option | Transform | Inverse | |---|---|---| | `none` | identity | identity (forecast clamped at zero) || `log` | `log(y)` | `exp( $\hat{y}$ )` || `seasonal_log_diff` | `log(yt) - log(y{t-12})` , the yearly growth (notebook 7) | `y{t-12} · exp( $\hat{z}_t$ )` |

TimesFM normalizes its input internally, so the transformations are not about scale; they test whether the model forecasts the **level**, the **log level** or the **growth rate** better. The quantiles are inverse-transformed the same way (all three are monotonic).

4. Covariates (past-and-future, next section).

```
In [5]: WINDOWS = {
    'all': None,
    **{f'last_{years}y': years for years in [14, 12, 10, 8, 6, 4]},
    'post_covid': POST_COVID_START,
}
COVID_ROWS = ['keep', 'remove', 'mask']
REMOVED = (pd.Timestamp('2020-01-31'), pd.Timestamp('2022-12-31')) # three full ye
MASKED = (COVID_START, pd.Timestamp('2022-12-31')) # pandemic and rebound
TARGET_TRANSFORMS = ['none', 'log', 'seasonal_log_diff']

def window_start(window, origin):
    spec = WINDOWS[window]
    if spec is None:
        return full_series.index[0]
    if isinstance(spec, pd.Timestamp):
        return spec
    return origin - pd.DateOffset(years=spec)

def window_exists(window, origin):
    spec = WINDOWS[window]
    return not isinstance(spec, pd.Timestamp) or origin > spec

def effective_covid_rows(covid_rows, origin):
    """Before the pandemic there is nothing to remove or mask."""
    return 'keep' if origin < REMOVED[0] else covid_rows

class SeasonalLogDiffTarget:
    """z_t = log(y_t) - log(y_{t-12}), positional; inverse from the last 12 months.

    def __init__(self, history, period=SEASONAL_PERIOD):
        self.period = period
        self.log_history = np.log(history.astype(float)).to_numpy()
        self.last_date = history.index[-1]
```

```

def transform(self, y):
    log_y = np.log(y.astype(float))
    return log_y - log_y.shift(self.period)

def inverse_transform(self, z, dates):
    dates = pd.DatetimeIndex(dates)
    steps = (dates.year - self.last_date.year) * 12 + (
        dates.month - self.last_date.month
    )
    assert ((steps >= 1) & (steps <= self.period)).all(), 'up to one season ahead'
    base = self.log_history[len(self.log_history) - self.period + steps - 1]
    # z may be (horizon,) or (horizon, n_quantiles)
    return np.exp(
        np.asarray(z, dtype=float) + base.reshape(-1, *((1] * (np.ndim(z) - 1))
    )

def make_context(window, covid_rows, transform, origin):
    """Context values (model scale, NaN where masked) and the fitted inverse."""
    history = full_series.loc[window_start(window, origin) : origin]
    if covid_rows == 'remove':
        history = history[~history.index.to_series().between(*REMOVED).to_numpy()]
    if transform == 'log':
        values = np.log(history.astype(float))
        inverse = lambda z, dates: np.exp(np.asarray(z, dtype=float)) # noqa: E731
    elif transform == 'seasonal_log_diff':
        target = SeasonalLogDiffTarget(history)
        values = target.transform(history).iloc[SEASONAL_PERIOD:]
        inverse = target.inverse_transform
    else:
        values = history.astype(float)
        inverse = lambda z, dates: np.asarray(z, dtype=float) # noqa: E731
    if covid_rows == 'mask':
        values = values.where(~values.index.to_series().between(*MASKED))
    return values, inverse

example_origin = post_covid_origins[0]
example = pd.DataFrame({
    covid_rows: make_context('last_6y', covid_rows, 'log', example_origin)[0]
    for covid_rows in COVID_ROWS
})
print(f'Context last_6y, log, origin {origin_label(example_origin)}, per COVID option')
display(example.notna().sum().rename('non-missing months'))
example.loc['2019-10-31':'2023-03-31'].round(2)

```

Context last_6y, log, origin 2024-08, per COVID option

keep 73

remove 37

mask 40

Name: non-missing months, dtype: int64

Out[5]:

	keep	remove	mask
date			
2019-10-31	12.93	12.93	12.93
2019-11-30	13.06	13.06	13.06
2019-12-31	13.39	13.39	13.39
2020-01-31	13.68	NaN	13.68
2020-02-29	13.61	NaN	13.61
2020-03-31	12.58	NaN	12.58
2020-04-30	6.33	NaN	NaN
2020-05-31	6.25	NaN	NaN
2020-06-30	6.48	NaN	NaN
2020-07-31	7.29	NaN	NaN
2020-08-31	9.02	NaN	NaN
2020-09-30	9.39	NaN	NaN
2020-10-31	10.12	NaN	NaN
2020-11-30	10.64	NaN	NaN
2020-12-31	11.29	NaN	NaN
2021-01-31	11.18	NaN	NaN
2021-02-28	10.78	NaN	NaN
2021-03-31	10.39	NaN	NaN
2021-04-30	9.82	NaN	NaN
2021-05-31	10.07	NaN	NaN
2021-06-30	10.09	NaN	NaN
2021-07-31	11.01	NaN	NaN
2021-08-31	10.61	NaN	NaN
2021-09-30	10.83	NaN	NaN
2021-10-31	11.45	NaN	NaN
2021-11-30	11.58	NaN	NaN
2021-12-31	12.07	NaN	NaN
2022-01-31	12.70	NaN	NaN
2022-02-28	12.65	NaN	NaN

	keep	remove	mask
date			
2022-03-31	12.53	NaN	NaN
2022-04-30	12.63	NaN	NaN
2022-05-31	12.22	NaN	NaN
2022-06-30	12.17	NaN	NaN
2022-07-31	12.58	NaN	NaN
2022-08-31	12.48	NaN	NaN
2022-09-30	12.52	NaN	NaN
2022-10-31	12.63	NaN	NaN
2022-11-30	12.77	NaN	NaN
2022-12-31	13.18	NaN	NaN
2023-01-31	13.79	13.79	13.79
2023-02-28	13.54	13.54	13.54
2023-03-31	13.27	13.27	13.27

3. Auxiliary Features (Past-and-Future Covariates)

TimesFM 3.0 reads numeric covariates alongside the target. A **past-and-future** covariate must be known over the context **and** the 12 forecast months, which is the case for features derived from the calendar:

Option	Covariates (one row each)
none	—
calendar	$\sin(2\pi \cdot \text{month}/12)$, $\cos(2\pi \cdot \text{month}/12)$ and the number of days in the month (centred)
covid	<code>is_covid</code> (April 2020 – December 2021) and <code>is_covid_recovery</code> (January – June 2022), the flags of notebooks 5–7
calendar+covid	both

- The **calendar** terms tell the model where each month sits in the year. A seasonal series already carries this in its own shape; the grid tests whether spelling it out helps.
- The **COVID flags** tell the model that the 2020–2022 values are exceptional, the same role as the exogenous regressors of SARIMAX (notebook 5). They only make sense with `keep` : with `remove` those months are gone and with `mask` they are missing, so the flags would carry no information. At the pre-COVID origins the flags are all zero, so

`covid` is the same as `none` and `calendar+covid` the same as `calendar` there (computed once).

Features derived from the target itself (lags, rolling means, growth) are not added: the model reads the whole target context and builds such summaries internally.

```
In [6]: COVARIATES = ['none', 'calendar', 'covid', 'calendar+covid']
        MEAN_DAYS_IN_MONTH = 365.25 / 12

def effective_covariates(covariates, origin):
    """Before the pandemic the COVID flags are all zero: drop them."""
    if origin >= COVID_START:
        return covariates
    return {'covid': 'none', 'calendar+covid': 'calendar'}.get(covariates, covariates)

def make_covariates(covariates, context_dates, future_dates):
    """Past-and-future covariates, shape (n_covariates, context + horizon), or None
    if covariates == 'none':
        return None
    dates = pd.DatetimeIndex(context_dates).append(pd.DatetimeIndex(future_dates))
    dates_series = dates.to_series()
    rows = []
    if 'calendar' in covariates:
        month = dates.month.to_numpy()
        rows += [
            np.sin(2 * np.pi * month / 12),
            np.cos(2 * np.pi * month / 12),
            dates.days_in_month.to_numpy() - MEAN_DAYS_IN_MONTH,
        ]
    if 'covid' in covariates:
        rows += [
            dates_series.between(COVID_START, COVID_END).to_numpy(),
            dates_series.between(POST_COVID_START, COVID_RECOVERY_END).to_numpy(),
        ]
    return np.asarray(rows, dtype=np.float32)

context, _ = make_context('all', 'keep', 'none', LATEST_ORIGIN)
future = val_actual_at(LATEST_ORIGIN).index
covariate_example = make_covariates('calendar+covid', context.index, future)
print(
    f'calendar+covid: {covariate_example.shape[0]} covariates x '
    f'{covariate_example.shape[1]} months '
    f'({len(context)} context + {len(future)} future)'
)
```

calendar+covid: 5 covariates x 440 months (428 context + 12 future)

4. Forecasting and Scoring One Configuration

`forecast_at` builds the context and the covariates of one origin, calls `model.predict(..., return_quantiles=True)` and inverse-transforms the median (the point forecast) and the 0.1 / 0.9 quantiles back to arrivals.

`score` is the scoring of notebooks 5, 7 and 8 (RMSE, MAE, MAPE, R^2 , **seasonal MASE**, Forecast Bias; MAE and Forecast Bias from the book's `calculate_metrics`), plus two interval metrics:

- **coverage 80%**: share of the validation months inside the 0.1–0.9 interval (ideal: 0.8).
- **interval width**: mean width of that interval, in % of the actual value.

The origins of a configuration are combined as in notebook 7: `MASE_mean` (the ranking metric), `MASE_worst` and `MASE_post`, plus the mean coverage and width. The seasonal MASE scale is the in-sample MAE of the seasonal naive on the origin's history, pandemic pairs (April 2020 – December 2022) left out, identical to the other notebooks.

```
In [7]: METRICS = ['RMSE', 'MAE', 'MAPE', 'R2', 'MASE', 'Forecast Bias']
AGG_METRICS = [
    'MASE_mean',
    'MASE_worst',
    'MASE_post',
    'coverage_80_mean',
    'interval_width_mean',
    'n_origins',
]
MASE_EXCLUDED = (COVID_START, pd.Timestamp('2022-12-31'))

def seasonal_mase_scale(history):
    """In-sample MAE of the seasonal naive (y_t - y_{t-12}), pandemic pairs exclude
    diffs = (history - history.shift(SEASONAL_PERIOD)).abs()
    in_pandemic = history.index.to_series().between(*MASE_EXCLUDED)
    touches = in_pandemic | in_pandemic.shift(SEASONAL_PERIOD, fill_value=False)
    return float(diffs[~touches.to_numpy()].mean())

MASE_SCALES = {
    origin: seasonal_mase_scale(full_series.loc[:origin]) for origin in ORIGINS
}

def score( # noqa: PLR0913, PLR0917
    actual, predicted, name, origin, low=None, high=None
):
    metrics = calculate_metrics(
        actual, predicted, name
    ) # book: MAE, MSE, Forecast Bias
    errors = (actual - predicted).to_numpy()
    result = {
        'RMSE': float(np.sqrt(metrics['MSE'])),
        'MAE': metrics['MAE'],
        'MAPE': float(np.mean(np.abs(errors / actual.to_numpy())) * 100),
```

```

        'R2': float(r2_score(actual.to_numpy(), predicted.to_numpy())),
        'MASE': float(metrics['MAE'] / MASE_SCALES[origin]),
        'Forecast Bias': metrics['Forecast Bias'],
        'abs_errors': np.abs(errors),
    }
    if low is not None:
        values = actual.to_numpy()
        result['coverage_80'] = float(np.mean((values >= low) & (values <= high)))
        result['interval_width'] = float(np.mean((high - low) / values) * 100)
    return result

def aggregate(per_origin):
    """Combine the per-origin scores of one configuration ({origin: scores})."""
    mase = pd.Series({origin: s['MASE'] for origin, s in per_origin.items()})
    post = mase[mase.index.isin(post_covid_origins)]
    latest = per_origin.get(LATEST_ORIGIN, {})

    def mean_of(key):
        values = [s[key] for s in per_origin.values() if key in s]
        return float(np.mean(values)) if values else np.nan

    return {
        'MASE_mean': float(mase.mean()),
        'MASE_worst': float(mase.max()),
        'MASE_post': float(post.mean()) if len(post) else np.nan,
        'coverage_80_mean': mean_of('coverage_80'),
        'interval_width_mean': mean_of('interval_width'),
        'n_origins': len(mase),
        # the latest origin forecasts the validation year (val.parquet)
        **{key: latest.get(key, np.nan) for key in METRICS},
    }

```

```

In [8]: class Config(NamedTuple):
    window: str
    covid_rows: str
    transform: str
    covariates: str
    symmetric: bool

    @property
    def label(self):
        return (
            f'{self.window} | {self.covid_rows} | {self.transform} | '
            f'{self.covariates} | {"symmetric" if self.symmetric else "plain"}'
        )

    def effective(config, origin):
        """The configuration run at `origin` (pre-COVID: no rows to drop, no flags)."""
        return config._replace(
            covid_rows=effective_covid_rows(config.covid_rows, origin),
            covariates=effective_covariates(config.covariates, origin),
        )

```

```

def forecast_at(config, origin):
    """Median, 0.1 and 0.9 quantile forecasts of the 12 months after `origin`."""
    context, inverse = make_context(
        config.window, config.covid_rows, config.transform, origin
    )
    future = val_actual_at(origin).index
    output = model.predict(
        context.to_numpy(dtype=np.float32),
        horizon=HORIZON,
        past_future_covariates=make_covariates(
            config.covariates, context.index, future
        ),
        return_quantiles=True,
        use_symmetric_averaging=config.symmetric,
        make_positive=config.transform == 'none',
    )
    median = inverse(output.forecast, future)
    quantiles = inverse(output.quantiles, future)
    return (
        pd.Series(median, index=future),
        quantiles[:, Q_LOW],
        quantiles[:, Q_HIGH],
    )

def evaluate_at(config, origin):
    result = {'origin': origin}
    start = time.perf_counter()
    try:
        predicted, low, high = forecast_at(config, origin)
        if not np.all(np.isfinite(predicted)):
            raise ValueError('non-finite forecast after the inverse transform')
        result.update(
            score(val_actual_at(origin), predicted, 'TimesFM', origin, low, high)
        )
        result.update({
            'prediction': predicted.to_numpy(),
            'low': low,
            'high': high,
            'error': None,
        })
    except Exception as exc: # noqa: BLE001 - reported in the results table
        result['error'] = f'{type(exc).__name__}: {exc}'[:250]
    result['fit_seconds'] = time.perf_counter() - start
    return result

def config_origins(config):
    return [origin for origin in ORIGINS if window_exists(config.window, origin)]

def summarize(config, origin_results):
    row = config._asdict()
    ok = {r['origin']: r for r in origin_results if r['error'] is None}
    errors = [r['error'] for r in origin_results if r['error'] is not None]
    row['error'] = errors[0] if errors else None

```

```

row['fit_seconds'] = sum(r['fit_seconds'] for r in origin_results)
row['origin_mase'] = {origin: r['MASE'] for origin, r in ok.items()}
if not errors:
    row.update(aggregate(ok))
latest = ok.get(LATEST_ORIGIN, {})
for key in ['prediction', 'low', 'high', 'abs_errors']:
    if key in latest:
        row[key] = latest[key]
return row

# the model out of the box: all history as it is, no transformation, no covariates
DEFAULT_CONFIG = Config('all', 'keep', 'none', 'none', False)
default_row = summarize(
    DEFAULT_CONFIG,
    [evaluate_at(DEFAULT_CONFIG, o) for o in config_origins(DEFAULT_CONFIG)],
)
print(f'Out of the box ({DEFAULT_CONFIG.label}):')
print({origin_label(o): round(v, 3) for o, v in default_row['origin_mase'].items()})
print(
    f'MASE_mean = {default_row["MASE_mean"]:.3f} | '
    f'80% coverage = {default_row["coverage_80_mean"]:.2f}'
)

```

Out of the box (all | keep | none | none | plain):
{'2017-08': 1.0, '2018-08': 0.618, '2023-08': 1.293, '2024-08': 3.949}
MASE_mean = 1.715 | 80% coverage = 0.79

5. Baselines

The references of notebooks 5, 7 and 8 at every origin: **naive** (the last training month repeated) and **seasonal naive** (the same month one year earlier). The model's out-of-the-box configuration above is a third reference: the grid is worth it only if a chosen context beats it.

```

In [9]: def baseline_forecasts_at(origin):
        history = full_series.loc[:origin]
        dates = val_actual_at(origin).index
        return {
            'naive': pd.Series(history.iloc[-1], index=dates),
            'seasonal_naive': pd.Series(
                history.reindex(dates - pd.offsets.MonthEnd(12)).to_numpy(), index=dates
            ),
        }

baseline_scores = {'naive': {}, 'seasonal_naive': {}}
for origin in ORIGINS:
    for name, predicted in baseline_forecasts_at(origin).items():
        baseline_scores[name][origin] = score(
            val_actual_at(origin), predicted, name, origin
        )

baselines = pd.DataFrame({

```

```

    name: aggregate(per_origin) for name, per_origin in baseline_scores.items()
}).T
BASELINE_MASE = float(baselines.loc['seasonal_naive', 'MASE_mean'])
baseline_forecasts = baseline_forecasts_at(LATEST_ORIGIN)

display(
    pd.DataFrame({
        name: {origin_label(o): s['MASE'] for o, s in per_origin.items()}
        for name, per_origin in baseline_scores.items()
    }).rename_axis('seasonal MASE by origin')
)
baselines[['MASE_mean', 'MASE_worst', 'MASE_post', *METRICS]].astype(float).round(3)

```

naive seasonal_naive

seasonal MASE by origin

2017-08	3.325103	0.680184
2018-08	2.441616	0.977196
2023-08	3.393431	1.462635
2024-08	6.148383	4.023995

Out[9]:

	MASE_mean	MASE_worst	MASE_post	RMSE	MAE	MAPE	I
naive	3.827	6.148	4.771	460440.495	319530.000	33.968	-0.9
seasonal_naive	1.786	4.024	2.743	255353.116	209126.083	26.478	0.4

6. Grid Search

Every valid combination of the five choices at every origin where it exists:

- `post_covid` only with `keep` ;
- the COVID flags (`covid` , `calendar+covid`) only with `keep` .

Each (configuration, origin) forecast runs once; the pre-COVID forecasts that are the same model (`remove` / `mask` = `keep` , COVID flags = none) are shared. About 360 configurations and 1,100 forecasts, a few minutes on the CPU.

```

In [10]: configs = [
    Config(*choices)
    for choices in itertools.product(
        WINDOWS, COVID_ROWS, TARGET_TRANSFORMS, COVARIATES, [False, True]
    )
    if not (choices[0] == 'post_covid' and choices[1] != 'keep')
    and not (choices[1] != 'keep' and 'covid' in choices[3])
]
tasks = list(
    dict.fromkeys(

```

```

        (effective(config, origin), origin)
        for config in configs
        for origin in config_origins(config)
    )
)
print(f'{len(configs)} configurations → {len(tasks)} forecasts')

grid_start = time.perf_counter()
with warnings.catch_warnings():
    warnings.simplefilter('ignore')
    by_task = {
        task: evaluate_at(*task) for task in tqdm(tasks, desc='TimesFM forecasts')
    }
print(f'Forecasting done in {time.perf_counter() - grid_start:.0f}s')

results = [
    summarize(
        config, [by_task[effective(config, o), o] for o in config_origins(config)]
    )
    for config in configs
]
results_df = pd.DataFrame(results)
failed = results_df[results_df['error'].notna()]
print(f'Failed: {len(failed)} / {len(results_df)}')
if len(failed):
    display(failed['error'].str.split(':').str[0].value_counts().rename('failures'))

```

360 configurations → 888 forecasts

TimesFM forecasts: 100%|██████████| 888/888 [19:58<00:00, 1.35s/it]

Forecasting done in 1,199s

Failed: 0 / 360

7. Log the Grid to MLflow

One run per configuration in the experiment **Brazil Tourism Forecasting TimesFM**: the five choices and the model as parameters; `MASE_mean` / `MASE_worst` / `MASE_post`, the validation-year metrics, the seasonal MASE of each origin (`MASE_origin_YYYY_MM`) and the interval metrics as metrics (`eval_split = validation`). No weights are trained, so no model artifact is logged.

```

In [11]: N_LOG_THREADS = 8
PARAM_KEYS = ['window', 'covid_rows', 'transform', 'covariates', 'symmetric']
ORIGIN_LABELS = ', '.join(origin_label(o) for o in ORIGINS)

def finite(value):
    return isinstance(value, (int, float, np.number)) and np.isfinite(value)

def log_result(client, result):
    params = {
        **{k: result[k] for k in PARAM_KEYS},
        'model': MODEL_ID,
    }

```

```

        'horizon': HORIZON,
        'train_end': f'{TRAIN_END:%Y-%m}',
        'eval_split': 'validation',
        'origins': ORIGIN_LABELS,
        'selection_metric': 'mean_seasonal_mase',
    }
    if result['error']:
        params['error'] = result['error']
    values = {
        **{key.replace(' ', '_'): result.get(key) for key in [*METRICS, *AGG_METRICS]},
        **{
            f'MASE_origin_{origin:%Y-%m}': value
            for origin, value in result['origin_mase'].items()
        },
        'fit_seconds': result['fit_seconds'],
    }
    timestamp = int(time.time() * 1000)
    metrics = [
        mlflow.entities.Metric(key, float(value), timestamp, 0)
        for key, value in values.items()
        if finite(value)
    ]
    run = client.create_run(
        experiment.experiment_id,
        run_name=' | '.join(str(result[k]) for k in PARAM_KEYS),
    )
    client.log_batch(
        run.info.run_id,
        metrics=metrics,
        params=[mlflow.entities.Param(k, str(v)) for k, v in params.items()],
    )
    client.set_terminated(
        run.info.run_id, status='FAILED' if result['error'] else 'FINISHED'
    )

client = mlflow.MlflowClient()
log_start = time.perf_counter()
with ThreadPoolExecutor(max_workers=N_LOG_THREADS) as pool:
    list(
        tqdm(
            pool.map(lambda r: log_result(client, r), results),
            total=len(results),
            desc='Logging to MLflow',
        )
    )
print(f'Logging done in {time.perf_counter() - log_start:,.0f}s')

```

```

Logging to MLflow: 100%|██████████| 360/360 [01:02<00:00, 5.76it/s]
Logging done in 63s

```

8. Results

All configurations that produced a forecast at every one of their origins, ranked by mean seasonal MASE. The seasonal naive is the bar to beat, and the out-of-the-box configuration shows what the choices of context add to the model as it comes.

```
In [12]: ranked = (
    results_df[results_df['error'].isna()]
    .sort_values('MASE_mean')
    .reset_index(drop=True)
)
ranked['beats_seasonal_naive'] = ranked['MASE_mean'] < BASELINE_MASE
default_rank = int(
    ranked.index[
        (ranked[PARAM_KEYS] == pd.Series(DEFAULT_CONFIG._asdict())).all(axis=1)
    ][0]
)
print(
    f'{len(ranked)} configurations ranked; '
    f'{ranked["beats_seasonal_naive"].sum()} beat the seasonal naive '
    f'(mean seasonal MASE {BASELINE_MASE:.3f})'
)
print(
    f'Out of the box: rank {default_rank + 1} of {len(ranked)} '
    f'(MASE_mean {ranked.loc[default_rank, "MASE_mean"]:.3f}); '
    f'best: {ranked.loc[0, "MASE_mean"]:.3f}'
)
ranked[[*PARAM_KEYS, *AGG_METRICS, 'RMSE', 'MAPE']].head(15).round(3)
```

360 configurations ranked; 224 beat the seasonal naive (mean seasonal MASE 1.786)
Out of the box: rank 178 of 360 (MASE_mean 1.715); best: 1.312

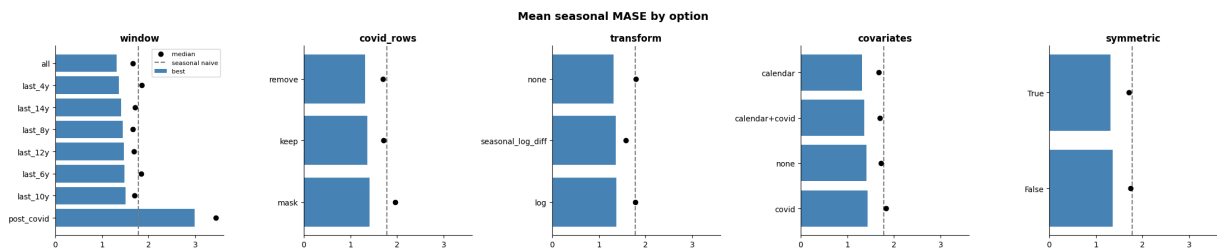
Out[12]:

	window	covid_rows	transform	covariates	symmetric	MASE_mean	MASE_v
0	all	remove	none	calendar	True	1.312	
1	last_4y	keep	seasonal_log_diff	calendar+covid	False	1.361	
2	last_4y	keep	seasonal_log_diff	calendar	False	1.367	
3	last_4y	keep	log	calendar+covid	True	1.377	
4	last_4y	keep	seasonal_log_diff	calendar+covid	True	1.378	
5	last_4y	keep	seasonal_log_diff	none	False	1.409	
6	last_4y	keep	seasonal_log_diff	calendar	True	1.410	
7	last_14y	mask	seasonal_log_diff	none	True	1.412	
8	last_4y	keep	log	none	False	1.420	
9	last_14y	mask	seasonal_log_diff	none	False	1.422	
10	last_14y	keep	seasonal_log_diff	covid	False	1.437	
11	last_14y	mask	seasonal_log_diff	calendar	False	1.441	
12	last_8y	mask	seasonal_log_diff	none	False	1.449	
13	last_14y	keep	seasonal_log_diff	calendar	False	1.454	
14	last_14y	keep	seasonal_log_diff	calendar+covid	False	1.455	

Effect of Each Choice

For every option of every choice, the best mean seasonal MASE among the configurations that use it (bars) and their median (dots). A choice matters when its options differ a lot; the dashed line is the seasonal naive.

```
In [13]: fig, axes = plt.subplots(1, len(PARAM_KEYS), figsize=(22, 4.5), sharex=True)
for ax, key in zip(axes, PARAM_KEYS):
    stats = ranked.groupby(key, sort=False)['MASE_mean'].agg(['min', 'median'])
    labels = [str(v) for v in stats.index]
    ax.barh(labels, stats['min'], color='steelblue', label='best')
    ax.scatter(stats['median'], labels, color='black', zorder=3, label='median')
    ax.axvline(BASELINE_MASE, color='grey', linestyle='--', label='seasonal naive')
    ax.set_title(key, fontweight='bold')
    ax.invert_yaxis()
    sns.despine(ax=ax)
axes[0].legend(fontsize=8)
fig.suptitle('Mean seasonal MASE by option', fontsize=14, fontweight='bold')
plt.tight_layout()
plt.show()
```



Top 10 Configurations — Validation Forecasts

The same view as notebooks 5, 7 and 8: the ten best configurations (by mean seasonal MASE), each with the last 24 training months, the actual validation year and the median forecast of the latest origin, now with the **80% interval** (0.1–0.9 quantiles) shaded.

```
In [14]: top10 = ranked.head(10)
train_plot = train_series.iloc[-24:]

fig, axes = plt.subplots(5, 2, figsize=(14, 26))
fig.suptitle(
    f'Top 10 TimesFM Configurations – Validation Set Forecasts '
    f'({val_series.index[0]:%b %Y}–{val_series.index[-1]:%b %Y})',
    fontsize=15,
    fontweight='bold',
)

for i, (ax, (_, row)) in enumerate(zip(axes.flat, top10.iterrows())):
    ax.plot(
        train_plot.index,
        train_plot.values,
        label='Train (last 24 m)',
        linewidth=1.5,
        color='slateblue',
        alpha=0.7,
    )
    ax.plot(
        val_series.index, val_series.values, label='Actual', linewidth=2, color='b1'
    )
    ax.fill_between(
        val_series.index,
        row['low'],
        row['high'],
        color='tomato',
        alpha=0.15,
        label='80% interval',
    )
    ax.plot(
        val_series.index,
        row['prediction'],
        label='Forecast (median)',
        linewidth=2,
        linestyle='--',
        color='tomato',
        marker='o',
        markersize=5,
    )
```

```

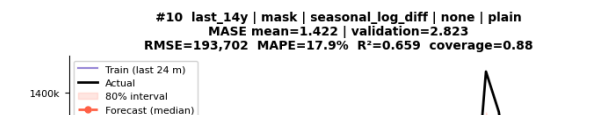
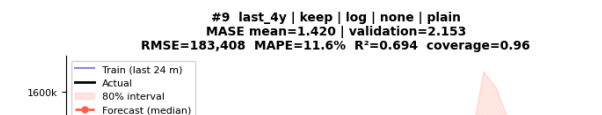
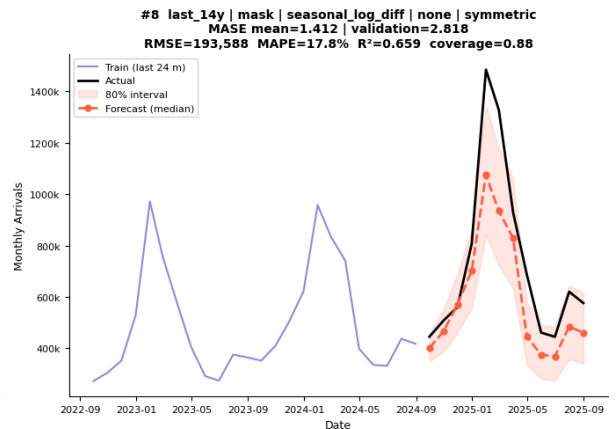
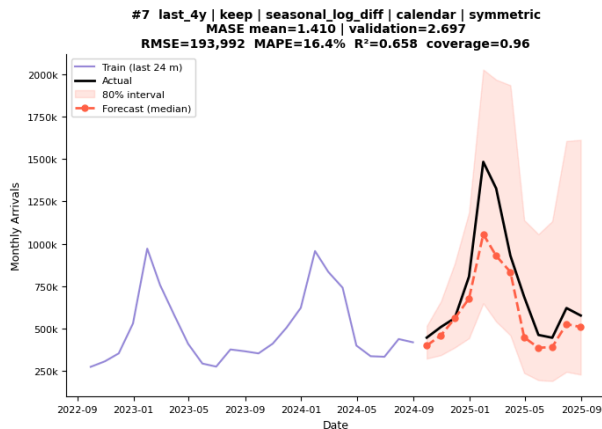
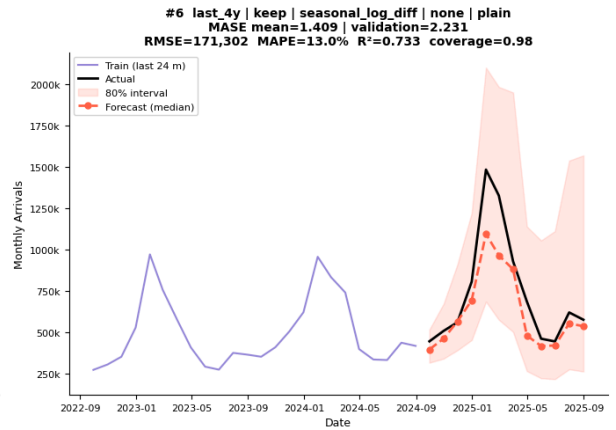
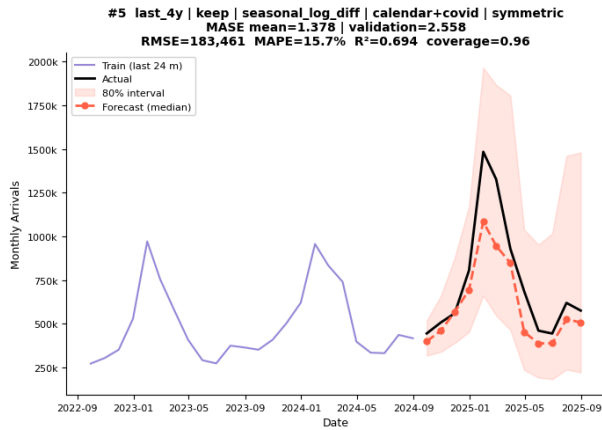
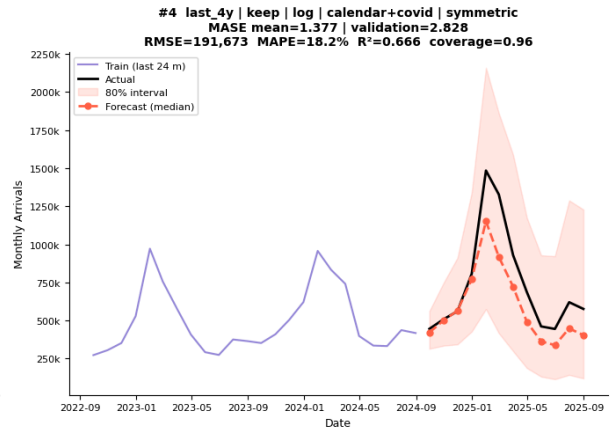
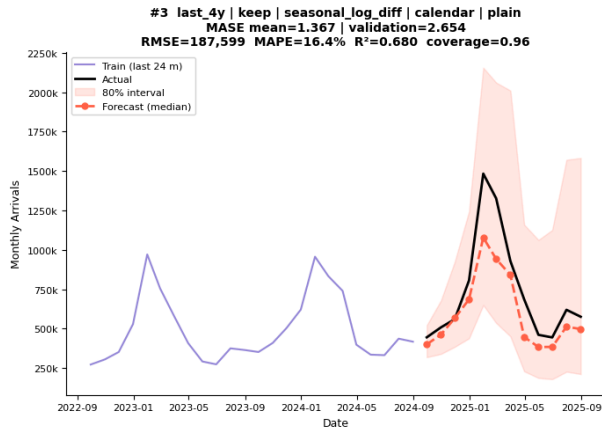
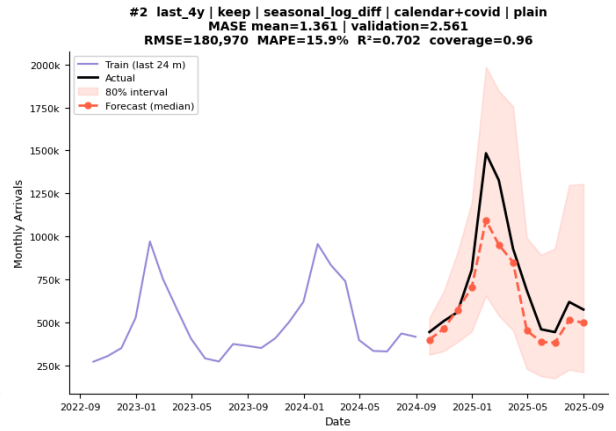
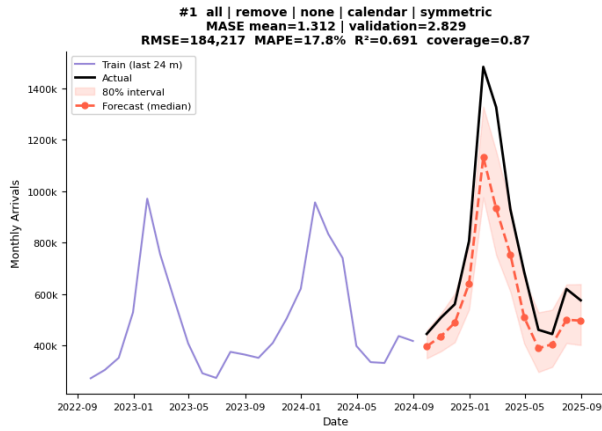
)
ax.set_title(
    f'#{i + 1} {Config(*row[PARAM_KEYS]).label}\n'
    f'MASE mean={row["MASE_mean"]:.3f} | validation={row["MASE"]:.3f}\n'
    f'RMSE={row["RMSE"]:.0f} MAPE={row["MAPE"]:.1f}% R²={row["R2"]:.3f} '
    f'coverage={row["coverage_80_mean"]:.2f}',
    fontsize=10,
    fontweight='bold',
)
ax.set_xlabel('Date', fontsize=9)
ax.set_ylabel('Monthly Arrivals', fontsize=9)
ax.yaxis.set_major_formatter(plt.FuncFormatter(lambda x, _: f'{int(x / 1000)}k')
ax.tick_params(axis='both', labelsize=8)
ax.legend(fontsize=8)
sns.despine(ax=ax)

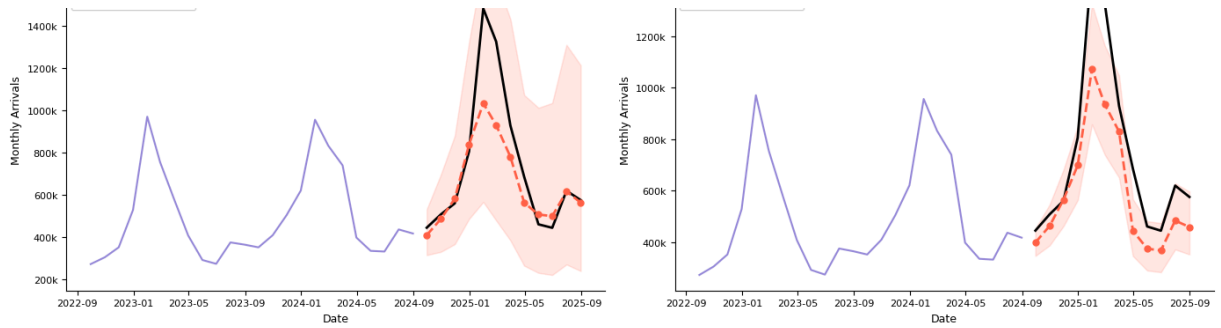
for ax in axes.flat[len(top10) :]:
    ax.set_visible(False)

plt.tight_layout(rect=[0, 0, 1, 0.97])
plt.show()

```

Top 10 TimesFM Configurations — Validation Set Forecasts (Sep 2024-Aug 2025)





9. Heatmaps of the Combinations

As in notebooks 5, 7 and 8, each cell summarizes all configurations that share the two choices on its axes with the **p95 of the mean seasonal MASE** (the value only the best 5% of the cell beat), with the number of configurations under it. The grid is small (about 5–60 configurations per cell), so the p95 is close to the best value of each cell.

```
In [15]: ORDERS = {
    'window': list(WINDOWS),
    'covid_rows': COVID_ROWS,
    'transform': TARGET_TRANSFORMS,
    'covariates': COVARIATES,
    'symmetric': [False, True],
}

HEATMAP_LAYOUT = [
    # (rows, columns, grid position)
    ('transform', 'covid_rows', (0, slice(0, 2))),
    ('covariates', 'transform', (0, slice(2, 4))),
    ('covariates', 'symmetric', (0, slice(4, 6))),
    ('window', 'covid_rows', (1, slice(0, 3))),
    ('window', 'transform', (1, slice(3, 6))),
]

MASE_LABEL_CAP = 10
P95_QUANTILE = 0.05 # p95: the value only the best 5% of a cell's configurations b

def p95(values):
    return values.quantile(P95_QUANTILE)

def mase_label(value, count):
    if pd.isna(value):
        return ''
    shown = f'>{MASE_LABEL_CAP}' if value > MASE_LABEL_CAP else f'{value:.2f}'
    return f'{shown}\n(n={int(count)})'

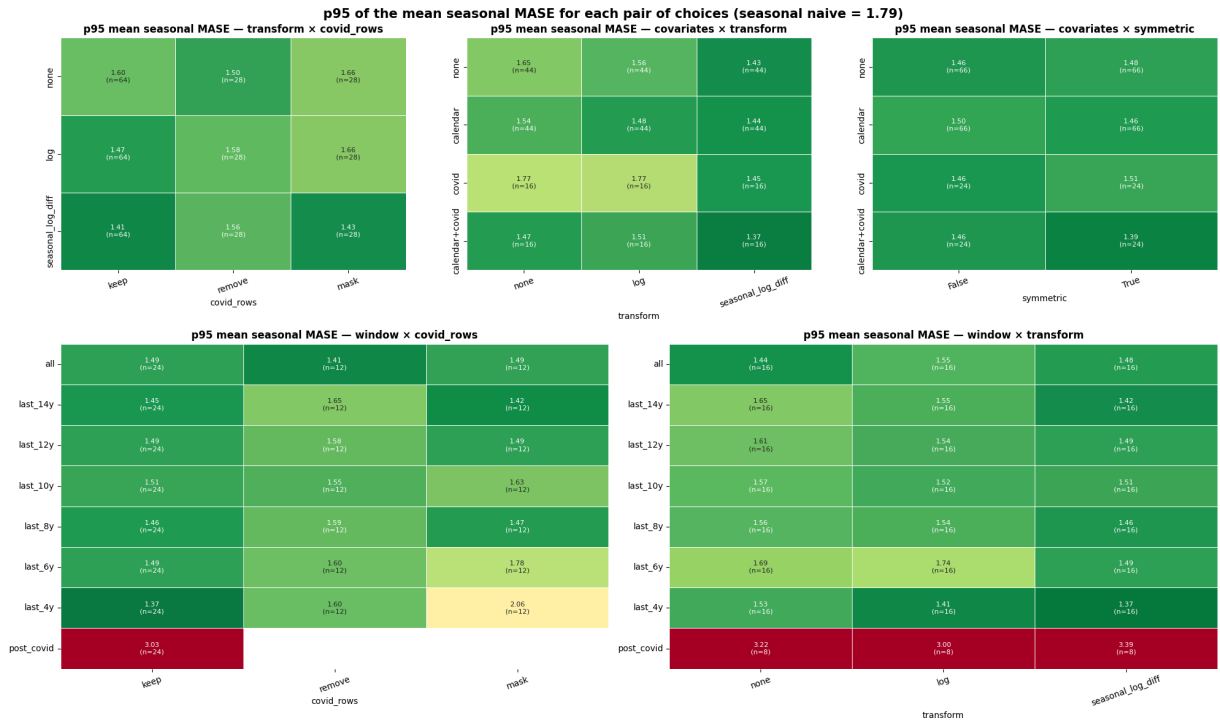
def p95_mase_pivot(rows, columns):
    """p95 of the mean seasonal MASE per cell, and the configurations per cell."""
    pivot, counts = (
        ranked.pivot_table(
            index=rows, columns=columns, values='MASE_mean', aggfunc=aggfunc
        ).reindex(
```

```

        index=[v for v in ORDERS[rows] if v in ranked[rows].unique()],
        columns=[v for v in ORDERS[columns] if v in ranked[columns].unique()],
    )
    for aggfunc in (p95, 'count')
)
labels = pd.DataFrame(
    [
        [mase_label(v, n) for v, n in zip(values, ns)]
        for values, ns in zip(pivot.to_numpy(), counts.to_numpy())
    ],
    index=pivot.index,
    columns=pivot.columns,
)
return pivot, labels

fig = plt.figure(figsize=(20, 12))
grid = fig.add_gridspec(2, 6, height_ratios=[1, 1.4])
for rows, columns, (r, c) in HEATMAP_LAYOUT:
    ax = fig.add_subplot(grid[r, c])
    pivot, labels = p95_mase_pivot(rows, columns)
    sns.heatmap(
        pivot,
        annot=labels,
        fmt='',
        annot_kws={'fontsize': 8},
        cmap='RdYlGn_r',
        vmin=ranked['MASE_mean'].min(),
        vmax=BASELINE_MASE * 1.5,
        linewidths=0.4,
        cbar=False,
        ax=ax,
    )
    ax.set_title(f'p95 mean seasonal MASE - {rows} × {columns}', fontweight='bold')
    ax.set_xlabel(columns)
    ax.set_ylabel('')
    ax.tick_params(axis='x', labelrotation=20)
fig.suptitle(
    'p95 of the mean seasonal MASE for each pair of choices '
    f'(seasonal naive = {BASELINE_MASE:.2f})',
    fontsize=15,
    fontweight='bold',
)
plt.tight_layout()
plt.show()

```



10. TimesFM vs the Trained Models

The best run of notebooks 5 (SARIMAX), 7 (local ML) and 8 (global ML), read from their MLflow experiments, next to the best and the out-of-the-box TimesFM configurations. All are scored on the same total, at the same origins, with the same seasonal MASE. A notebook whose runs are missing, or were logged with other origins, is reported instead of compared.

```
In [16]: TRAINED_EXPERIMENTS = {
    'SARIMAX (notebook 5)': ('Brazil Tourism Forecasting ARIMA', 'mase'),
    'Local ML (notebook 7)': ('Brazil Tourism Forecasting ML', 'MASE'),
    'Global ML (notebook 8)': ('Brazil Tourism Forecasting ML Global', 'MASE'),
}

def best_run(experiment_name, prefix):
    found = mlflow.get_experiment_by_name(experiment_name)
    if found is None:
        return None
    finished = "attributes.status = 'FINISHED' and params.eval_split = 'validation'"
    runs = client.search_runs(
        [found.experiment_id],
        filter_string=finished,
        order_by=[f'metrics.{prefix}_mean ASC'],
        max_results=1,
    )
    return runs[0] if runs else None

def row_of(approach, configuration, source):
    return {
        'approach': approach,
```

```

        'configuration': configuration,
        'MASE_mean': source['MASE_mean'],
        'MASE_worst': source['MASE_worst'],
        'MASE (validation year)': source['MASE'],
        'RMSE (validation year)': source['RMSE'],
        'MAPE (validation year)': source['MAPE'],
    }

comparison = []
for approach, (experiment_name, prefix) in TRAINED_EXPERIMENTS.items():
    run = best_run(experiment_name, prefix)
    if run is None:
        print(f'{approach}: no runs in "{experiment_name}"')
        continue
    if run.data.params.get('origins') != ORIGIN_LABELS:
        print(
            f'{approach}: logged with other origins ({run.data.params.get("origins")})
        )
        continue
    m = run.data.metrics
    comparison.append(
        row_of(
            approach,
            run.info.run_name,
            {
                'MASE_mean': m.get(f'{prefix}_mean'),
                'MASE_worst': m.get(f'{prefix}_worst'),
                'MASE': m.get(prefix),
                'RMSE': m.get('rmse', m.get('RMSE')),
                'MAPE': m.get('mape', m.get('MAPE')),
            },
        )
    )
best = ranked.iloc[0]
comparison.append(
    row_of('TimesFM, best context', Config(*best[PARAM_KEYS]).label, best)
)
comparison.append(
    row_of('TimesFM, out of the box', DEFAULT_CONFIG.label, ranked.loc[default_rank])
)
comparison.append(
    row_of(
        'Seasonal naive',
        'same month last year',
        baselines.loc['seasonal_naive'],
    )
)
pd.DataFrame(comparison).set_index('approach').round(3)

```

Global ML (notebook 8): no runs in "Brazil Tourism Forecasting ML Global"

Out[16]:

	configuration	MASE_mean	MASE_worst	MASE (validation year)	RMSE (validation year)	MAPE (validation year)
approach						
SARIMAX (notebook 5)	BEST SARIMAX(0,1,2) (0,1,1)[12] (last_4y_covid_...	0.992	1.258	1.258	103693.084	7.302
Local ML (notebook 7)	xgboost seasonal_log_diff last_4y keep ...	1.188	1.448	1.448	94163.485	9.729
TimesFM, best context	all remove none calendar symmetric	1.312	2.829	2.829	184216.923	17.815
TimesFM, out of the box	all keep none none plain	1.715	3.949	3.949	262799.177	24.614
Seasonal naive	same month last year	1.786	4.024	4.024	255353.116	26.478

Reading the Results

Question	Where
Does TimesFM beat the seasonal naive without any training?	<code>beats_seasonal_naive</code> , section 8
How much do the context choices add to the model out of the box?	Rank of the out-of-the-box configuration, section 8 and 10
How much history should the model read?	<i>Effect of each choice</i> (<code>window</code>) and heatmap <i>window × covid_rows</i>
Is it better to keep, remove or mask the pandemic?	Heatmap <i>transform × covid_rows</i>
Level, log level or growth rate?	Heatmap <i>transform × covid_rows</i> and <i>window × transform</i>
Do calendar terms or COVID flags help?	Heatmap <i>covariates × transform</i>
Are the intervals trustworthy?	<code>coverage_80_mean</code> (ideal 0.8) and the shaded bands of the top 10
Does a zero-shot foundation model beat the trained models?	Section 10

Things to keep in mind:

- **Selection on validation data.** The best of about 360 context configurations is optimistic, although much less than the thousands of trained configurations of notebooks 5 and 7. The unbiased score comes from the test year in the final notebook.
- **Pretraining data.** TimesFM was pretrained on a large public corpus that may include tourism series. Zero-shot results on the validation years are not a sign of leakage from this project, but part of the pattern knowledge comes from outside it.
- **License.** TimesFM 3.0 weights are for research and non-commercial use only.